

On the Motion of a Rigid Body in an Incompressible Fluid Medium.

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It appears that up to now, for no case, however simple, has the resistance which a rigid body moved through a fluid at rest experiences from that fluid been derived from the general equations of hydrodynamics known since Euler; or, what amounts essentially to the same thing, there is no example of a purely theoretical determination of the modifications which an immovable rigid body situated in the interior of a fluid produces in the progressive motion of that fluid. A notable mathematician, very practiced in investigations of this kind,² has therefore been led to the opinion that, for the problem just mentioned, even in the case where the fluid is of infinite extent and the rigid body is given the simplest shape, the known integration methods do not suffice. This opinion, however, is unfounded, and the problem can be treated completely when the rigid body has the shape of a sphere, or also that of an ellipsoid. Of these two cases only the first, as the simpler one, will be discussed in this notice.

To treat the problem first in the second of the forms mentioned above, let c be the radius of the immovable sphere and let its center be the origin of rectangular axes x, y, z . Let an accelerating force σ act on the initially resting homogeneous fluid, whose density is denoted by ϱ , and suppose that at the same time t this force has everywhere the same intensity and direction, but may change arbitrarily with time, so that its components α, β, γ are given functions of t . If p denotes the pressure occurring at the point (x, y, z) after the time t , and u, v, w the three components of the velocity, set, for abbreviation,

$$r = \sqrt{x^2 + y^2 + z^2},$$

$$\lambda = \int_0^t \alpha dt, \quad \mu = \int_0^t \beta dt, \quad \nu = \int_0^t \gamma dt,$$

$$\varphi = \left(1 + \frac{c^3}{2r^3}\right)(\lambda x + \mu y + \nu z).$$

Then, for the determination of p, u, v, w , one has

$$u = \frac{\partial \varphi}{\partial x} = \left(1 + \frac{c^3}{2r^3}\right)\lambda - \frac{3c^3 x}{2r^5}(\lambda x + \mu y + \nu z),$$

$$v = \frac{\partial \varphi}{\partial y} = \left(1 + \frac{c^3}{2r^3}\right)\mu - \frac{3c^3 y}{2r^5}(\lambda x + \mu y + \nu z),$$

¹In the 1852 volume of the Academy reports, the communication is introduced on p. 12 with the words: Herr Lejeune Dirichlet read on some cases in which the motion of a rigid body in an incompressible fluid medium can be determined theoretically. K.

²Résumé des leçons de Mécanique données à l'École Polytechnique by M. Navier, p. 480.

$$w = \frac{\partial \varphi}{\partial z} = \left(1 + \frac{c^3}{2r^3}\right) \nu - \frac{3c^3 z}{2r^5} (\lambda x + \mu y + \nu z),$$

$$\frac{1}{\varrho} p = T - \frac{c^3}{2r^3} (\alpha x + \beta y + \gamma z) - \frac{1}{2} (u^2 + v^2 + w^2),$$

where T denotes an arbitrary quantity depending only on t , which, as Euler already observed, remains undetermined in every problem of hydrodynamics so long as the pressure at a point of the surface of the fluid is not prescribed for every instant. This indeterminacy in the expression for p plainly has no influence on the result obtained when all elementary pressure forces acting on a closed surface are combined into a single force and a couple. For the surface of our sphere, all these elementary forces have a single resultant, passing through the center; and for its components one finds, by a very simple calculation, the values

$$\frac{2\pi}{3} c^3 \varrho \alpha, \quad \frac{2\pi}{3} c^3 \varrho \beta, \quad \frac{2\pi}{3} c^3 \varrho \gamma.$$

Thus for the resultant one obtains the expression

$$\frac{2\pi}{3} c^3 \varrho \sigma,$$

and at the same time it is clear that it is parallel to the force σ . Hence in the present case the pressure exerted by the moving fluid on the surface of the rigid body depends only on the accelerating force acting at each instant, and not on the velocity of the fluid. If the accelerating force ceases to act, the pressure also vanishes, λ, μ, ν become constant, and the motion immediately becomes permanent, so that u, v, w no longer depend on t . The permanent motion which then occurs is always of the same kind, if one takes account only of the ratios of the velocities and, for the paths described by the fluid particles, only of the relative position of these curves and not their absolute position in space.

All these curves are plane, and all their planes always pass through a fixed straight line, which may be regarded as the axis of the system of curves. Around this axis the curves lie symmetrically: the curves lying in one plane generate all the others by rotation about the axis. The direction of the axis, which passes through the center of the sphere, is determined by the three angles it forms with the coordinate axes, and the cosines of these angles are equal to

$$\frac{\lambda}{\sqrt{\lambda^2 + \mu^2 + \nu^2}}, \quad \frac{\mu}{\sqrt{\lambda^2 + \mu^2 + \nu^2}}, \quad \frac{\nu}{\sqrt{\lambda^2 + \mu^2 + \nu^2}},$$

where by λ, μ, ν one is to understand the values of these quantities after they have become constant. The equations of the curves just mentioned take a very simple form if polar coordinates are introduced. Draw any plane through the axis, and let θ denote the angle between the axis and the radius vector r drawn in that plane to the point in question. Then all the curves are contained in the equation

$$(r^3 - c^3) \sin^2 \theta = \varepsilon r,$$

where the parameter ε must run through the interval from 0 to ∞ in order that the equation represent all curves lying in the plane. One sees without difficulty that for large ε these curves more and more assume the shape of straight lines parallel to the axis, while as ε decreases the form of such a curve approaches ever more closely a semicircle, which continues at its two endpoints along the prolonged diameter.

To derive from the results just indicated the simplest case of the motion of a rigid body in a resting fluid, imagine the sphere homogeneous, or at least suppose that its center of gravity coincides with its center. Let ϱ' denote the mean density of the sphere, and let an accelerating force act on all parts of it whose components are

$$-\frac{\varrho}{2\varrho'} \alpha, \quad -\frac{\varrho}{2\varrho'} \beta, \quad -\frac{\varrho}{2\varrho'} \gamma.$$

Then the pressure exerted by the fluid is cancelled at every instant, and the sphere may be regarded as movable without ceasing to be at rest and without modifying the motion of the fluid determined above. Now let the accelerating force with components $-\alpha, -\beta, -\gamma$ be imagined as acting everywhere on the whole mechanical system formed by the sphere and the fluid, which has the character of a so-called free system. By composition one obtains the motion arising in our initially resting system for the case where an accelerating force acts on the sphere whose components have the values

$$-\left(1 + \frac{\varrho}{2\varrho'}\right)\alpha, \quad -\left(1 + \frac{\varrho}{2\varrho'}\right)\beta, \quad -\left(1 + \frac{\varrho}{2\varrho'}\right)\gamma,$$

while the fluid is driven by no force. In this motion the components of the velocity at time t for all parts of the sphere are $-\lambda, -\mu, -\nu$, whereas in the fluid, at the same time and at the point determined by the coordinates

$$x - \int_0^t \lambda dt, \quad y - \int_0^t \mu dt, \quad z - \int_0^t \nu dt,$$

the velocity components have the values

$$u - \lambda, \quad v - \mu, \quad w - \nu.$$

Thus the sphere, under the action of the accelerating force $\left(1 + \frac{\varrho}{2\varrho'}\right)\sigma$, arbitrarily variable in direction and intensity, moves in the resisting medium exactly as it would move in empty space if, at each instant, the force were diminished, with its direction retained, in the ratio of $2\varrho' + \varrho$ to $2\varrho'$. The additional force $\frac{\varrho}{2\varrho'}\sigma$ is therefore expended in overcoming the resistance of the fluid. But this resistance does not correspond to the usual conception of the action of a fluid medium on a rigid body moving in it, according to which a resistance is already present and must be overcome even if the motion occurring at one instant is not to be altered during the next element of time. In contrast, in our case the motion of the rigid body immediately becomes rectilinear and uniform as soon as the accelerating force ceases to act. The resistance here does not depend at all on the existing motion, but only on the change of motion to be produced in the next element of time, and it is always equal to the same aliquot part of the force that would produce the same change in empty space, directed exactly opposite to it—a fact which one could hardly have expected in the case where the velocity is to be diminished.